Jinhao Jing

jinhaojing@link.cuhk.edu.cn | tomie56.github.io | github.com/Tomie56 | Google Scholar

Education

The Chinese University of Hong Kong, Shenzhen — Sep 2024 – Mar 2027 (expected)
M.S. in Artificial Intelligence and Robotics — *Shenzhen, China*
GPA 3.78/4.0, ranked 7/117. Advised by Prof. Benyou Wang.

Shanghai Jiao Tong University — Sep 2019 – Jun 2024
B.Eng. in Mechanical Engineering — *Shanghai, China*
GPA 3.3/4.3.

Research Experience

Tsinghua University — Apr 2026 – Present
Visiting Student, THU Metaverse AI Lab; with Prof. Yuwang Wang — *Beijing, China*

- Research on representation engineering and model steering, including activation steering for RL exploration [P2].

IQuest Lab, Ubiquant — Jun 2026 – Present
Pre-training Data Algorithm Intern; with Ran Tao and Bryan Dai — *Beijing, China*

- Led the training of general quality scoring models for pre-training, and the WebCode pre-training corpus (∼1.2T tokens) for **IQuest-Q1**, an open-source 320B MoE model for agentic coding, reasoning and tool use [Code] [Model].

SenseTime Research — Jul 2025 – Apr 2026
Research Algorithm Intern; with Zheng Ma and Lewei Lu — *Shenzhen, China*

- Built mid-training data for the **SenseNova V6.5** multimodal model: ∼2M cross-disciplinary and geometric reasoning samples (∼2B tokens) through cleaning, augmentation, distilling and synthesis. Ran data ablations on a 30B base model to guide mid-training data selection; One-Step-Train [C2] was adopted in the production data pipeline.

Publications

* denotes equal contribution.

Conference and Journal Papers

[C1] **GeoSym127K: Scalable Symbolically-verifiable Synthesis for Multimodal Geometric Reasoning**
Jinhao Jing*, Zheng Ma*, Jinwei Liang*, Qiannian Zhao, Shawn Chen, Jing Yang, Por Lip Yee, Prayag Tiwari, Jingjing Bai, Benyou Wang, Lewei Lu, Zhan Su
NeurIPS 2026, Evaluations and Datasets Track. **Oral (15 of 3,757 valid submissions).** [arXiv] [Code] [Data]

[C2] **Efficient Data Selection for Multimodal Models via Incremental Optimization Utility**
Jinhao Jing*, Qiannian Zhao*, Chao Huang, Zhan Su
ACM International Conference on Multimedia (ACM MM) 2026. [arXiv]

[C3] **Steve-Evolving: Open-World Embodied Self-Evolution via Fine-Grained Diagnosis and Dual-Track Knowledge Distillation**
Zhengwei Xie, Zhisheng Chen, Ziyan Weng, Jinhan Li, Chenglong Li, Zikai Xiao, Jingwei Song, **Jinhao Jing**, Vireo Zhang, Kun Wang
NeurIPS 2026. [arXiv]

[C4] **CP Merging: Joint LoRA Merging using Canonical Polyadic Decomposition**
Zhan Su, Fengran Mo, Guojun Liang, Jinghan Zhang, **Jinhao Jing**, Bingbing Wen, Prayag Tiwari, Jian-Yun Nie
Transactions on Machine Learning Research (TMLR), 2026. [OpenReview]

Preprints and Manuscripts

[P1] **Predictive Memory Localization: Forecasting Selective Intervention Paths from Internal Signals**
Jinhao Jing, Tian Zeyu, Lucas Qingyang Fang, Zhisheng Chen, Shuang Chen, Yuhao Luo, Qiannian Zhao
arXiv preprint, 2026. [arXiv] [Code] [Data]

[P2] **Exploration via Activation-Space Steering for Reinforcement Learning with Verifiable Rewards**
Yunan Zeng, **Jinhao Jing**, Xi Cheng, Yuwang Wang
arXiv preprint, 2026. [arXiv]

[P3] **Know What You Know: Metacognitive Entropy Calibration for Verifiable RL Reasoning**
Qiannian Zhao*, Chen Yang*, **Jinhao Jing***, Yunke Zhang, Xuhui Ren, Lu Yu, Shijie Zhang, Hongzhi Yin
arXiv preprint, 2026. [arXiv]

Research Interests

- **Training-data value.** How data, training and model capability relate. Prior signals judge a sample before training; posterior ones measure what training on it changed. Mainstream practice stays on the prior side and scores **data-likeness** — resemblance to text already deemed good. I would rather attribute capability back to the data that produced it, and work out how signals of both kinds should be related, to select training data.
- **Verifiable data synthesis and evaluation.** A judge model's verdict is an opinion; a solver's can be rerun. Building training data, rewards and benchmarks whose correctness is decided by symbolic solvers and executable rules — mostly for multimodal reasoning and constrained decision-making.

More broadly, I am interested in reinforcement learning with verifiable rewards and in model-internal signals — uncertainty, activations, intervention sites — as a source of supervision.

Research Highlights

Training-Data Value — *Can a sample's worth to a model be known before training, or only through it?*

- **One-Step-Train [C2].** Proposed **marginal training utility**: the loss change after one simulated update on a lightweight multimodal proxy (InternVL3-1B). On 350K math CoT samples, the top 20% selected by the proxy transfers to a 30B in-house model and Qwen3-VL-8B, beating Full-SFT by **+8.8** and LLM-as-a-Judge selection by **+5.6** points at 43% lower training cost.
- **Quality-scorer Model.** Distilling an LLM's judgments into a scorer concentrates three problems. *Sampling*: score-based selection compresses correlated dimensions and strips diversity, so the pool is drawn for quality and coverage at once. *Annotation*: a 1–5 scale cannot separate documents inside a band, so a second pass splits each grade and hard cases are grown by active learning and LLM rewriting. *Loss*: the closer two labels sit, the noisier their order and the cheaper a regression error — ranking and regression cannot be weighted alike across the gap. The resulting 8- and 20-head scorers reach **86.7%** adjacent-pair win rate on a 100K held-out set and now filter the corpora above.
- **RubricFlow.** Built a **preference DAG** over LLM comparisons to induce local rubrics from chains sharing anchors. Over 19.7K blind pairs, the induced within-band order holds at **68.8%** against **55.8%** for the underlying coarse grades.
- **Combining signals.** Likeness scores are cheap but blind; influence functions, loss trajectories and gradient alignment do see the training effect, at a cost that rules out whole-corpus use; sparse autoencoders give a third reading, naming the features a sample carries instead of scoring it. The question I care about is composition — which signal to trust at corpus scale, which to spend at the decision boundary, and what their disagreement is itself evidence of.

Verifiable Data Synthesis and Evaluation — *How can correctness be guaranteed rather than assumed?*

- **GeoSym127K [C1].** Multimodal models hallucinate on geometry diagrams and answer-verified training data is scarce, so I built a **neuro-symbolic synthesis engine**: geometric relations live in SymPy, diagram topology evolves under a type-conditioned grammar, and an analytic solver verifies every answer before it is kept. 51K images, 127K answer-verified QA pairs and 55K solver-verified CoT traces (2K+ monthly downloads); SFT and RL on GeoSym improve Qwen3-VL-8B by **+22.21** on MathVerse (Vision-Only).
- **GeoSym-Bench [C1].** The solver written to generate also grades, diagnoses and constrains. 511 expert-curated problems scored by answer and symbolic-equivalence checks, on which GeoSym training lifts Qwen3-VL-8B from 17.42% to **26.81%** — within reach of Claude Opus 4.6 (27.79%). The same checks now drive reasoning-error diagnosis (GeoSym-Debug), an extension to solid geometry (GeoSym3D), and a geometry agent whose parsing, solving and tool calls are **verifier-gated** (GeoClaw).
- **YGO-Bench.** Competitive card games are an unusually good test for language models: the rules are intricate and fully executable, legal moves are combinatorial, and winning needs planning under hidden information rather than recall. As a Yu-Gi-Oh! player I knew where the hard cases live, so I built a benchmark from tournament decks and game states with every label checked by a rule engine, testing rule understanding and constrained decision-making in LLMs and agents. [Site]

Model-Internal Signals — *What does a model reveal beyond whether its answer was right?*

- **Entropy-calibrated RLVR [P3].** A verifiable reward cannot separate a confident solution from a lucky guess, and a rollout group that is uniformly right or wrong yields no advantage at all. A zero-overhead token-entropy proxy with **asymmetric calibration** recovers a within-group gradient without touching the verifier.
- **Activation-Space Exploration [P2].** Confidence describes the paths a policy sampled, not the ones it never tried. Clustering activation differences between correct and incorrect responses to the same problem gives a **bank of latent exploration vectors** steering later rollouts elsewhere; at equal budget, mean@8 over six math benchmarks exceeds GRPO by **+4.87** (Qwen3-1.7B-Base) and **+1.87** (Qwen3-8B-Base).
- **Predictive Memory Localization [P1].** Knowing where a fact is stored does not say whether that site has a usable operating range. Treating an intervention path (record, direction, layer, strength) as a **forecast target**, the response to a very weak probe predicts which paths move the target without damaging semantic neighbours at untried strengths.